

YIBO WANG

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≡ EDUCATION

-
- Purdue University, Department of Computer Science** **Aug. 2025 – Present**
PhD in Computer Science *West Lafayette, IN*
◦ GPA: 4.00/4
- Sichuan University, College of Computer Science** **Sept. 2021 – Jun. 2025**
B.Eng in Computer Science *Sichuan, China*
◦ GPA: 3.85/4

≡ PUBLICATIONS

Water: A Workload-Adaptive Knob Tuning System

- **Yibo Wang**, Jiale Lao, Chen Zhang, Cehua Yang, Yuanchun Zhou, Jianguo Wang, Mingjie Tang
- Under Revision at **VLDB 2027**, Paper, Project

Can You Trust the Vectors in Your Vector Database? Black-Hole Attack from Embedding Space Defects

- Hanxi Li, Jianan Zhou, Jiale Lao, **Yibo Wang**, Zhengmao Ye, Yang Cao, Junfen Wang, Mingjie Tang
- Under revision at **VLDB 2027**, Paper Project

QUITE: A Query Rewrite System Beyond Rules with LLM Agents

- Yuyang Song, Hanxu Yan, Jiale Lao, **Yibo Wang**, Yufei Li, Yuanchun Zhou, Jianguo Wang, Mingjie Tang
- **Preprint**, Paper Project

GPTuner: An LLM-Based Database Tuning System

- Jiale Lao, **Yibo Wang**, Yufei Li, Jianping Wang, Yunjia Zhang, Zhiyuan Chen, Wanghu Chen, Mingjie Tang, Jianguo Wang
- **SIGMOD Record 2025**, Paper

GPTuner: A Manual-Reading Database Tuning System via GPT-Guided Bayesian Optimization

- Jiale Lao, **Yibo Wang**, Yufei Li, Jianping Wang, Yunjia Zhang, Zhiyuan Chen, Wanghu Chen, Mingjie Tang, Jianguo Wang
- **VLDB 2024**, Paper, Project
- Selected as **SIGMOD Research Highlight**

A Demonstration of GPTuner: A GPT-Based Manual-Reading Database Tuning System

- Jiale Lao, **Yibo Wang (Co-first)**, Yufei Li, Jianping Wang, Yunjia Zhang, Zhiyuan Chen, Wanghu Chen, Yuanchun Zhou, Mingjie Tang, Jianguo Wang
- **SIGMOD 2024 Demo**, Paper, Video

≡ Research Experience

Research Assistant – Purdue University **Oct. 2024 – Present**

Advisor: Prof. Jianguo Wang

- Automatic Optimization of DiskANN and Relational Databases with Machine Learning and Large Language Model

Research Assistant – Sichuan University **July. 2023 – June. 2025**

AI and System Lab. Advised by Prof. Mingjie Tang

- Automatic Optimization of Relational Databases with Machine Learning and Large Language Model

☰ Awards

- SIGMOD Research Highlight, 2024

☰ Services

- Program Committee CIKM [2026]
- Reviewer - TKDE [2025]
- Reviewer - Information Sciences [2025]

☰ Teaching

- Teaching Assistant, Purdue CS348: Information Systems [Fall 2025][Spring 2026]